# How Much Does a Parity Experiment Prove?

Poorva Barve
poorvabarve@gmail.com

***Abstract*—Harbor ports more than eighty benchmarks through adapters. Each port undergoes a parity experiment comparing the original harness and the port with the same agent and model, usually three times each. The port passes when the run-score ranges overlap. With three runs per side and exchangeable scores under parity, the rule has a false-alarm rate of 10 percent without ties and a lower rate with ties. Adding runs reduces the rule's chance of detecting a fixed gap. Under common normal noise, more noise makes a port with that gap more likely to pass. I re-analyse 58 published comparisons with three runs per side. Their median absolute gap is 0.7 points, but equivalence within 3 points can be shown for only 22 comparisons and within 5 points for 36. The rule passes the four largest gaps, up to 15.6 points. At the median observed standard deviation and a true gap of zero, an equivalence test needs about seven runs per side for a 3-point margin and 80 percent power. Per-task outcomes expose task-level disagreement in 14 of 27 adapters. Among the tests evaluated in binary-outcome simulations at a 5 percent false-alarm rate, a stratified-by-task test detects the most injected defects with three runs. I provide a five-step protocol and a verdict tool.**

***Index Terms*—agent evaluation, benchmark portability, equivalence testing, reproducibility**

## I. Introduction

Two adapters in the Harbor repository illustrate the problem. For USACO the original harness scored 39.5, 40.5 and 41.0 in three runs and the Harbor port scored 45.7, 42.1 and 41.1. The ranges do not overlap, by 0.1 point, so under the matching criterion of Harbor's contributor guide [1] the port does not match. For RexBench (12 tasks) the original harness scored 33.33 in every run and the port scored 16.67, 58.33 and 50.0. The ranges overlap, so the port passes, with an 8.3-point gap between the means. The same rule gives opposite verdicts, and the comparison it rejects is the one whose data say more about closeness.

Harbor is the evaluation framework behind Terminal-Bench [2]; existing benchmarks are ported to it through adapters, and the maintainers' cross-benchmark comparisons and their curated Harbor-Index task set are built on the ports [3]. A port that drifts is a different benchmark under the same name. The contributor guide therefore requires a parity experiment and states the acceptance rule in one sentence: the two sides match if and only if their run-score ranges overlap [1]. The maintainers' paper words the check as the means agreeing "within a margin consistent with their standard errors" [3].

I ask three questions. RQ1: what does the range-overlap rule guarantee? RQ2: what do the published runs establish? RQ3: what would a better check cost? My contributions are the rule's error analysis, an audit under a documented metric-selection rule, and run-count and per-task guidance supported by defect simulations. I use public parity files, per-task outcomes and the maintainers' paper; no benchmark was rerun.[1] A failed comparison does not establish an adapter bug; noise in agent evaluations is already documented [4]–[6].

## II. Background: Harbor parity experiments

An adapter lets Harbor run an existing benchmark. Before it is merged it goes through code review and a parity experiment: its author runs the same agent and model on the original harness and on the port, $k$ times each with $k = 3$ by default, and reports each side as mean $\pm$ sample standard error of the mean (SEM) [3]; the guide requires at least two runs per side and prefers three or more [1]. The guide asks that the two sides' scores be "statistically indistinguishable" and states the matching criterion as range overlap: $\max(A) \geq \min(B)$ and $\max(B) \geq \min(A)$ for the two lists of run scores $A$ and $B$. Non-overlap is to be treated as a systematic adaptation error until evidence indicates otherwise, and the parity step is complete when runs on both sides overlap and any remaining gap is documented, with evidence, as sampling noise [1]. The paper states the check twice in other words: the means "agree within a margin consistent with their standard errors" (D.4.2) and the port "must confirm statistical equivalence with the original benchmark within sample SEM" (D.7) [3].

The snapshot is Harbor commit `dc933be` (10 October 2026). Its 84 parity files hold 121 experiments. Eighty-two compare an original harness with its port; 61 have three runs on both sides, and 58 remain after dropping two whose run lists are identical on both sides and one whose metric is not on a 0-to-100 scale. These 58 form the main analysis set. The range-overlap sentence entered the guide on 19 April 2026; 51 of the 58 records predate it, when the guide asked only that averages be comparable. I apply the current rule to all because it is the rule stated today, so the results describe the rule, not past reviews. From the Hugging Face dataset I recovered per-task result files for 49 adapters, 44 with both sides [7].

## III. RQ1: What the rule guarantees

**Closed form.** Two sides produce $k$ independent run scores each. Under a perfect port the $2k$ scores are exchangeable, so every ordering is equally likely. The ranges fail to overlap exactly when all $k$ scores of one side exceed all $k$ of the other, which happens in $2(k!)^2$ of the $(2k)!$ orderings, so $\Pr(\text{mismatch} \mid \text{perfect port}) = 2(k!)^2/(2k)!$, that is $1/3$, $1/10$, $1/35$ and $1/126$ at $k = 2, 3, 4, 5$, for any continuous noise

[1]Data, scripts, analysis plan with dated amendments, per-task parsers, number-to-script map and verdict tool: https://github.com/barveP/benchmark-port-parity-archive.

TABLE I
RULES AT $k = 3$ UNDER NORMAL NOISE: FALSE ALARM, DETECTION OF 1 SD AND 2 SD GAPS, AND DETECTION OF A $t$-STATISTIC AT THE SAME FALSE ALARM (FA).

| rule | false alarm | 1 SD | 2 SD | $t$ at same FA |
|---|---|---|---|---|
| range overlap (Harbor) | 0.10 | 0.26 | 0.62 | 0.64 |
| SEM bars overlap | 0.25 | 0.50 | 0.86 | 0.86 |
| gap within SE of difference | 0.37 | 0.63 | 0.92 | 0.92 |
| gap within one SEM | 0.44 | 0.68 | 0.94 | 0.94 |
| pooled $t$-test, 5% | 0.05 | 0.16 | 0.46 | 0.46 |
| Welch $t$-test, 5% | 0.03 | 0.12 | 0.37 | 0.38 |

distribution. When scores are pass counts that can tie, ties count as overlap and the rate is lower: 4 to 9% at $k = 3$ over a grid of 10 to 1,000 tasks and base rates 0.1 to 0.5.

**Other rules.** Table I compares six rules at $k = 3$ under normal noise with a common run-to-run standard deviation (SD): range overlap; mean $\pm$ SEM bars overlap (S1) and gap within the standard error of the difference (S2), the two readings of the paper's D.4.2 wording; gap within the larger of the two SEMs (S3), the literal reading of D.7; and pooled and Welch $t$-tests at 5%. The last column gives the detection of a $t$-statistic tuned to each rule's own false-alarm rate. S2 and the pooled $t$-test are one statistic at different thresholds, and S1, S3 and the Welch test are close to it; range overlap is a different statistic, but at its own false-alarm rate it loses only two points of detection (0.62 against 0.64). Inefficiency is not its problem; its problems are structural.

**Three properties.** First, for scores that cannot tie the false-alarm rate is set by $k$ alone and cannot be chosen (Fig. 1a). Second, for any fixed gap the rule's detection falls as runs are added, because the no-overlap event for $k+1$ runs per side is contained in the event for $k$ runs; for a 2 SD gap it falls from 0.78 at $k = 2$ to 0.09 at $k = 10$, while a $t$-test's rises from 0.22 to 0.99 (Fig. 1b). More evidence makes the check weaker. Third, noise helps a port pass. For a true gap of 2 points and the same noise on both sides, the pass probability at $k = 3$ rises from 0.02 at an SD of 0.5 points to 0.68 at the published runs' median SD of 1.7 points and towards 0.90 as the noise grows (Fig. 1c). Repeating the analysis with run scores built from the per-task pass probabilities of 30 adapters (Section V) leaves the ordering unchanged; with ties the overlap rule's median false alarm is 0.07.

**What a pass means.** With three runs per side and exchangeable scores under parity, a pass is a failure to reject with a difference test whose false-alarm rate is at most 10%. It never means that the port is within any stated distance of the original: under a perfect port with $k = 3$ the smallest margin within which equivalence can be shown is 2.3 run SDs on average, which is 3.9 points at the median SD of 1.7. Read literally, the paper's wordings are stricter than the guide's rule: they flag 25 to 44% of perfect ports (Table I).

## IV. RQ2: WHAT THE PUBLISHED RUNS ESTABLISH

**Method.** I built one comparison per experiment under a rule written down before the analysis: which metric row is the main one (the row matching the paper's tables, else the headline metric, else the first point-scale row), which rows are excluded, and how units are handled. Six amendments were logged with date and reason; two refined the rule before any count was computed, and the one made after the first counts left the within-5 count at 36 of 58. For each comparison I computed the gap (Harbor minus original), the pooled-$t$ 90% confidence interval, and the *smallest provable margin*: the distance from zero to the farther end of that interval. Equivalence within $\pm m$ is shown when the whole 90% interval lies inside $\pm m$, the two-one-sided-tests (TOST) procedure at the 5% level [8], [9].

**Results.** The ports look close. Over the 58 comparisons the median absolute gap is 0.67 points, the median run-to-run SD 1.71 points, and no gap is distinguishable from zero at the 95% level. The evidence that they are close is thin: the quartiles of the smallest provable margin are 2.4, 4.4 and 6.0 points. Equivalence within 2, 3 and 5 points can be shown for 7, 22 and 36 of the 58, and one in four cannot show it even within 6 points (Fig. 2).

**The rule's verdicts.** Fifty-seven comparisons pass range overlap; 21 of them cannot show equivalence within 5 points and 35 cannot within 3. The rule passes the four largest gaps: Frontier-CS at $-15.6$ points (a 10-task subset with a run SD of 16 points), RexBench at $+8.3$, PIXIU at $-4.9$ and GDB at $+4.6$, whose smallest provable margins are 44.2, 35.5, 20.3 and 11.6 points. The one failure, USACO, misses overlap by 0.1 point and has a smallest provable margin of 5.8 points, which 18 of the 57 passes exceed. The paper's SEM wordings would have failed 4, 12 and 18 of the 58 under readings S1, S2 and S3.

**Selection.** If all 58 ports were perfect and runs independent, range overlap would flag 5.8 of them on average for scores that cannot tie, or 3.9 at the tie-adjusted rate of the per-task model; it flagged one, and the chance of one or fewer is 0.017 and 0.09 under the two rates. The debugging workflow can select for passing records: non-overlap is treated as an error to investigate and rerun. This is a possible selection mechanism, not a reconstruction of every adapter's review history. Published pass rates therefore cannot validate the rule, and the rates of Section III are properties of the rule under stated noise models, not of the published ports.

**Stability.** Counts at fixed margins are knife-edge (a 0.1-point shift moves the within-5 count between 34 and 37; Welch intervals give 31), so the quartiles are the stable summary. Subset comparisons (22 of the 58) are noisier than full-set ones (median run SD 2.3 against 1.5 points).

## V. RQ3: A BETTER CHECK AND ITS COST

**Runs.** Table II gives the runs per side an equivalence test needs for 80% power (an 80% chance of showing equivalence when the true gap is as stated) at the SD quartiles observed in the 58 comparisons (1.1, 1.7 and 2.7 points), for margins

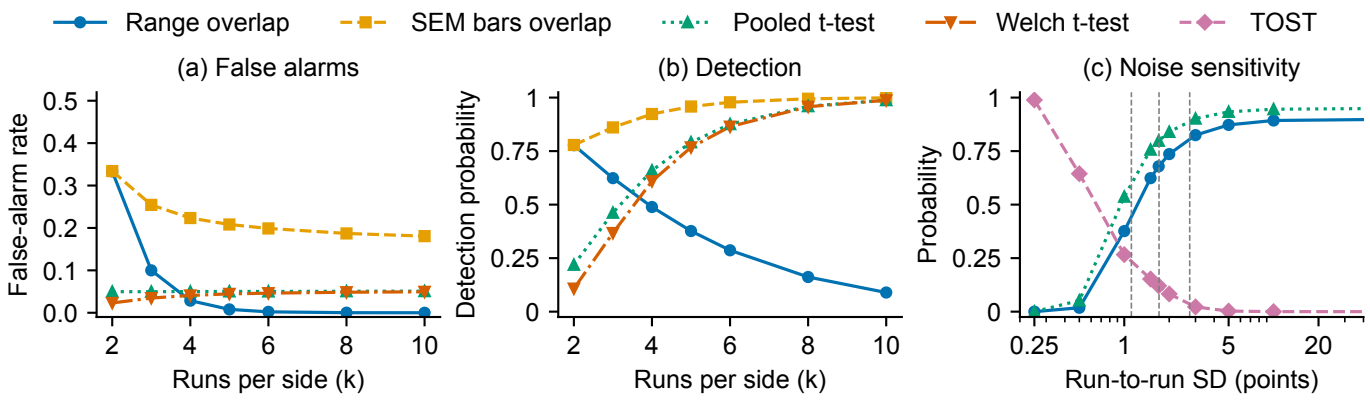



Fig. 1. Normal run-to-run noise. (a) False-alarm rate and (b) detection of a 2 SD gap as runs per side ($k$) increase. Both $t$-tests use the 5% level. (c) For a true 2-point gap at $k = 3$, the curves show the probabilities that range overlap passes, the pooled $t$-test fails to reject, and TOST (Section IV) shows equivalence within 3 points. Vertical lines mark the observed SD quartiles.

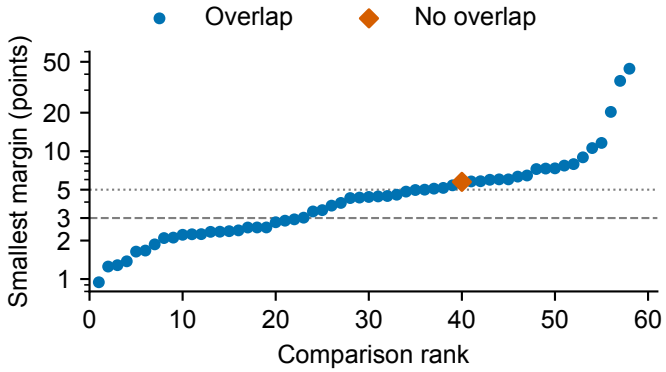



Fig. 2. Smallest equivalence margin supported by each of the 58 comparisons with three runs per side (TOST, 90% interval). The dashed and dotted lines mark 3 and 5 points, respectively. The diamond marks USACO at rank 40. The four largest gaps (Frontier-CS, RexBench, PIXIU, and GDB) all pass range overlap.

TABLE II
RUNS PER SIDE FOR 80% POWER OF THE POOLED-$t$ TOST (5% LEVEL, NORMAL NOISE, COMMON SD) TO SHOW EQUIVALENCE WITHIN $\pm m$ POINTS AT THE OBSERVED SD QUARTILES, FOR A TRUE GAP OF 0 (AND OF 0.6 POINTS).

| margin | SD 1.1 | SD 1.7 | SD 2.7 |
|---|---|---|---|
| $\pm 2$ | 7 (9) | 14 (20) | 33 (49) |
| $\pm 3$ | 4 (4) | 7 (8) | 16 (18) |
| $\pm 5$ | 3 (3) | 3 (4) | 6 (7) |

of 2, 3 and 5 points, under a true gap of 0 and of 0.6 points (about the observed median). At the median noise a 3-point margin needs about 7 runs per side; per comparison with its own SD the median need is 7 (quartiles 4 and 15.5), and two comparisons would need more than 400. The plug-in rule, the smallest $k$ whose interval from the observed SD just fits the margin, has only 12 to 78% power across this table.

**Per-task outcomes.** The published files hold per-task results for both sides in 44 adapters [7], and the guide's debugging playbook, used when ranges fail to overlap, already asks for a per-task split and treats tasks that are constant on one side but flip on the other as a sign of a systematic issue [1]; I propose that check for every port. Of the 44, 27 meet the inclusion rule (replicate runs, at least 20 shared tasks, files that reproduce the published run scores or carry a documented exception). Pairing the sides task by task does not tighten the interval for the gap. Under shared per-task pass probabilities the paired interval should be about a sixth narrower than the run-level one, only because the latter must estimate its noise from three numbers; in fact it is wider in 15 of the 26 adapters with a run-level interval (SWE-bench has one original run). Two things account for this. In 14 of 27 adapters the two sides disagree on individual tasks more than shared pass probabilities allow, even though a within-task permutation test does not reject the mean gap for any of the 27 ($p > 0.05$) and a stratified-by-task test (Cochran-Mantel-Haenszel, comparing the sides within each task and pooling across tasks) rejects for none of the 21 adapters with binary outcomes. One pair of sides cannot say whether this comes from the port or from side-specific effects such as seeds or caching. And in several adapters the run-level interval is narrow for a reason the per-task data do not support: in 9 of the 30 adapters with binary outcomes and validated replicate Harbor runs, the between-run variance of the published Harbor run scores is below a tenth of what their own within-task flip rates imply under independent outcomes (I call this ratio the dispersion ratio; quartiles 0.10, 0.52, 1.26). With three runs a ratio this low arises by chance in about one adapter in ten, so 2.6 of the 30 would be expected; 9 is unlikely ($p = 0.0007$). This diagnostic is consistent with dependence within or between runs, but does not identify its cause; I report it as an open data-quality question.

**Injected defects.** I simulated both sides from the per-task pass probabilities of the same 30 adapters, with a defect that breaks (always fails) or leaks (always passes) a fraction of tasks on the port side, 2,000 replications per cell (Fig. 3). With three runs, a defect that breaks 10% of tasks (a loss of 5.4 points on average) is caught by range overlap 0.64

of the time, at a false-alarm rate of 0.07 under this model; by the pooled $t$-test 0.59; by the stratified test 0.67; and by a paired $t$-test on per-task differences 0.54, the last three at 0.05. The paired $t$ suffers because its standard error absorbs the between-task spread of the defect itself, which does not shrink with runs. With ten runs the pooled $t$ and the stratified test catch the same defect 0.86 of the time, the paired $t$ 0.67, and range overlap 0.36. A perfect port is shown equivalent within 5 points 0.51 of the time at $k = 3$, 0.75 at $k = 5$ and 0.94 at $k = 10$. The 21 adapters with a dispersion ratio of at least 0.1 give the same picture; a flat Beta(1,1) prior lowers detection at $k = 3$ to 0.43, 0.34, 0.45 and 0.39 (same order) and the within-5 rate for a perfect port to 0.20; resampling the 8 fractional-score adapters' observed rewards, where the stratified test does not apply, gives 0.72, 0.67 and 0.59 for range overlap, pooled $t$ and paired $t$.

**A verdict tool.** The verdict script reads any `parity_experiment.json` and prints the overlap verdict, the gap with its 90% interval, the smallest provable margin, and the runs per side needed for 80% power at that row's own noise if the true gap were 0. For USACO: gap $+2.6$ ($-0.5$ to $+5.8$), margin 5.8 points, 7 runs for $\pm 3$. For RexBench: gap $+8.3$ ($-18.8$ to $+35.5$), margin 35.5 points, 464 runs.

## VI. Recommended protocol

1) State the margin before running: how many points of difference would make the port a different benchmark (3 and 5 points are illustrative; justify the margin for the benchmark).
2) Choose the run count from the noise for 80% power at that margin (Table II); when the noise is unknown, start with at least five runs per side and fix the total from their SD before looking at the gap. If the runs needed are out of reach (464 per side for RexBench at $\pm 3$), enlarge the parity set where a subset was used, or report the smallest provable margin instead.
3) Report the gap with its 90% interval and the smallest provable margin. Declare "equivalent within $\pm m$" only when the interval lies inside $\pm m$; otherwise "not yet shown", which asks for a larger experiment, not a rewrite.
4) Publish aligned per-task outcomes for both sides, report a stratified-by-task test (for binary outcomes; the pooled $t$ on run scores otherwise), and investigate tasks that differ in every run, which the guide's playbook already treats as a sign of a systematic issue [1].
5) Keep range overlap as a sanity check at most.

The cost is about seven runs per side instead of three at the median observed noise for a 3-point margin, and a reporting change.

## VII. Threats to validity

*Noise models.* Run scores are treated as independent draws with a common variance; binomial and per-task models rank the rules the same way. Run-level shocks such as infrastructure noise [6] are not modelled. *Selection.* Published records may be selected by fixes and reruns (Section IV), so their pass rates cannot validate any rule; the reported rates are model properties. *Few runs.* Three runs give a noisy SD; per-task data give a second estimate. *Metric rule.* The main-metric rule and its amendments are a choice; Section IV reports how the within-5 count moves under a 0.1-point shift and under Welch intervals. Mixed metric types sit on one point scale; subsets are noisier than full sets. *Per-task data.* One format per adapter, validated only against the published run scores; 13 of the 44 two-sided folders are excluded, 4 are Terminal-Bench legs reported in the repository only, and 6 of the 27 used carry documented exceptions. The per-task model's prior changes detection rates by up to a factor of two, and its independence assumption understates run-level rates for the under-dispersed adapters. *Snapshot.* One commit of a moving repository; no benchmark was rerun.

## VIII. Related work

That overlap of error bars or intervals is a poorly calibrated test is classical [10], [11], and researchers misjudge it even so [12]. Equivalence testing with a stated margin is standard in other fields [8], [9]. Noise in language-model and agent evaluation has been quantified with error bars and clustering [4], variance decomposition [5] and controlled infrastructure experiments [6]; re-implementations of the same benchmark are known to drift [13], and other ecosystems accept re-implementations with a fixed tolerance [14] or a test against a reference distribution [15]. Recent work applies paired per-task tests and TOST at a stated margin to agent scaffolds and compressed models [16], [17], while a leaderboard audit uses paired tests and states no equivalence margin [18]; Zhang *et al.* make the same gate-versus-certificate point [19]. Inside Harbor, the maintainers' paper reports a Welch $t$-test and a permutation test for two adapters [3], and one port that had passed parity with one model later proved to leak gold tests and was fixed the same day [20]. I focus on the acceptance rule itself, the evidence in the published parity runs, and the run budget needed for an equivalence check.

## IX. Conclusion

With three runs per side and exchangeable scores under parity, range overlap has a false-alarm rate of at most 10%; a pass does not certify closeness. Of the 58 published comparisons, 36 show equivalence within 5 points. For a 3-point margin at the median observed SD and a true gap of zero, about seven runs per side provide 80% power. Stated margins, equivalence tests and aligned per-task outcomes make the check more informative.

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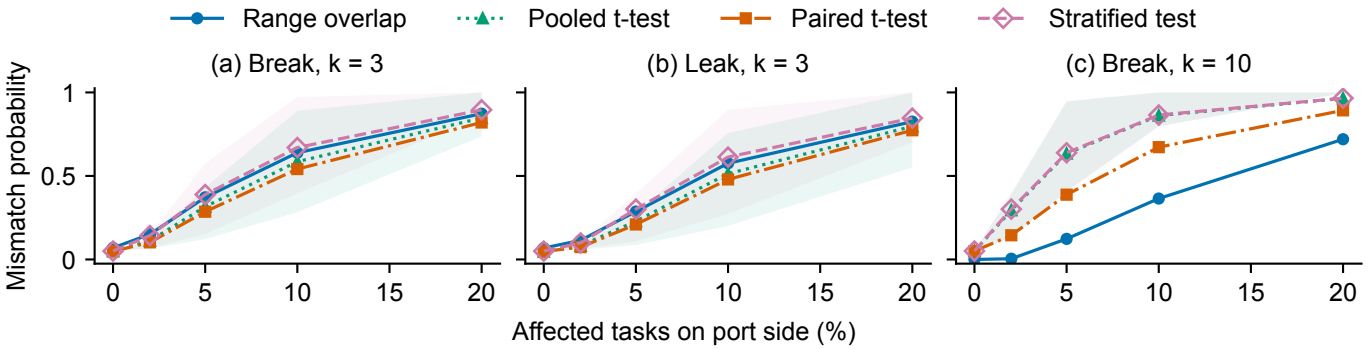



Fig. 3. Detection of injected defects, mean over 30 adapters simulated from their per-task outcomes: (a) break defects at $k = 3$, (b) leak defects at $k = 3$, (c) break defects at $k = 10$. Bands: interquartile range across adapters for the pooled $t$ and stratified tests; in (c) their curves coincide. The value at zero is the false-alarm rate.

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